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VLSI - 第八周作业.

1. 已知 $Q^* = JQ' + KQ$

驱动方程

$$\begin{cases} J_0 = K_0 = 1 \\ J_1 = \overline{Q_3} \cdot \overline{Q_2} \cdot Q_0' & K_1 = Q_0' \\ J_2 = Q_3 \cdot Q_0' & K_2 = Q_1' Q_0' \\ J_3 = Q_2' Q_1' Q_0' & K_3 = Q_0' \end{cases}$$

状态方程

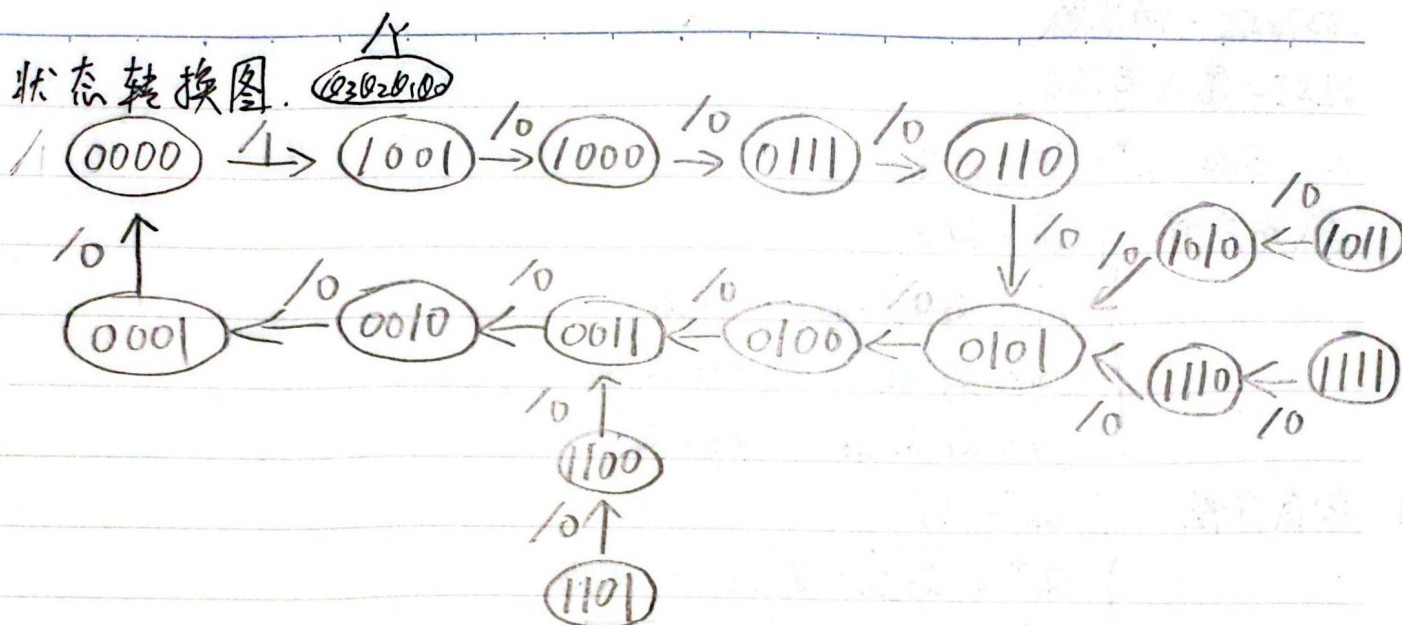
$$\begin{cases} Q_0^* = Q_0' \\ Q_1^* = \overline{Q_3} \overline{Q_2} \cdot Q_1' Q_0' + Q_1 Q_0 \\ Q_2^* = Q_3 Q_2' Q_0' + Q_2 \overline{Q_1} \overline{Q_0}' \\ Q_3^* = Q_3' Q_2' Q_1' Q_0' + Q_3 Q_0 \end{cases}$$

输出方程.

$$y = Q_3' Q_2' Q_1' Q_0'$$

状态转换表

Q_3	Q_2	Q_1	Q_0	Q_3^*	Q_2^*	Q_1^*	Q_0^*	Q_3	Q_2	Q_1	Q_0	Q_3^*	Q_2^*	Q_1^*	Q_0^*
✓ 0	0	0	0	1	0	0	1	1	0	1	1	1	0	1	0
✓ 0	0	0	1	0	0	0	0	1	1	0	1	1	1	0	0
✓ 0	0	1	0	0	0	0	1	1	1	1	0	0	1	0	1
✓ 0	1	0	0	0	0	1	1	1	1	1	1	1	1	1	0
✓ 1	0	0	0	0	1	1	1								
✓ 0	0	1	1	0	0	1	0								
✓ 0	1	0	1	0	1	0	0								
✓ 1	0	0	1	1	0	0	0								
✓ 0	1	1	0	0	1	0	1								
1	0	1	0	0	1	0	1								
1	1	0	0	0	0	1	1								
✓ 0	1	1	1	0	1	1	0								



综上, 可以自启动.

2. 由图可知 预置数 $D_3D_2D_1D_0 = 0011 = 3$.

预置端 $LD = Q_3Q_0$. 满足的最小值为 $Q_3Q_2Q_1Q_0 = 1001 = 9$

∴ 70160 同步置零计数.

∴ "1" 会被计入.

3 - 9 : 七进制.

3. always @ (posedge clk or posedge reset) begin
if (reset) begin

count <= 4'b0001

高 → 低

④ 对应 $Q_3Q_2Q_1Q_0$

end else begin

count <= {count[2:0], ~count[3]}.

① 非阻塞

③ 左高位 右低

end

end